Problem Statement 2 – Output Guidelines

Package: MC011A – PTCA (Percutaneous Transluminal Coronary Angioplasty)

Purpose of the Outputs

Participants must generate two outputs for each claim:

S. No.	Output
1	**Structured, row-wise output capturing:** • Objective interpretation of pre-procedure and post-procedure angiographic images • Correlation with written angiography report • Alignment with PTCA STG requirements • Stage and timeline awareness
2	**Textual summary** that consolidates all structured outputs into a human-readable note suitable for PPD / CPD review, without making diagnostic, approval, or payment decisions

📌 *Note:* For PTCA, both pre-procedure and post-procedure angiograms are mandatory.

SECTION 1: Claim Details ***(1 column)***

Claim No.

- **Description:** Unique identifier for the claim
- **Type:** String / Integer
- **Purpose:** Links angiographic interpretation to a specific PTCA claim

SECTION 2: Image Classification ***(2 columns)***

1. **Type of Radiological Image – Pre-Procedure Stage**
    - **Description:** Free text, Imaging submitted at pre-authorization / diagnostic stage
    - **Examples:**
        - "Cardiac angiogram (single still)"
        - "Cardiac angiogram (multiple stills)"
        - "Absent"
    - **Purpose:** Confirms diagnostic angiogram availability as per PTCA STG

2. **Type of Radiological Image – Post-Procedure Stage**
    - **Description: Imaging submitted at claim stage after PTCA**
    - **Expected Values:**
        - **"Cardiac angiogram (single still)"**
        - **"Cardiac angiogram (multiple stills)"**
        - **"Absent"**
    - **Purpose:** Confirms post-procedure evidence submission (mandatory in PTCA)

SECTION 3: Image Description – Coronary Anatomy ***(9 columns)***

✅ Describe what is seen in each vessel, not what it means

These fields capture **objective, observable image features described under following nine mandatory headings/ columns**

3A. Pre-Procedure Image Description ***(4 columns, pipe separated sub-categories) -***

1. **LMCA** (Left Main Coronary Artery)
    - Proximal
    - Distal
2. **LAD** (Left Anterior Descending artery)
    - Proximal
    - Mid
    - Distal
3. **LCX** (Left Circumflex Artery)
    - Proximal

- Distal
- OM

4. **RCA** (Right Coronary Artery)
    - Proximal
    - Mid
    - Distal

- Expected Content (per artery): free-text, observational description such as -
    - "Proximal – no disease"
    - "Mid – 40–50% luminal narrowing/ stenosis"
    - "Distal – not visualised"

3B. Post-Procedure Image Description ***(5 columns, pipe separated sub-categories) -***

1. **LMCA** (Left Main Coronary Artery)
    - Proximal
    - Distal
2. **LAD** (Left Anterior Descending artery)
    - Proximal
    - Mid
    - Distal
3. **LCX** (Left Circumflex Artery)
    - Proximal
    - Distal
    - OM
4. **RCA** (Right Coronary Artery)
    - Proximal
    - Mid
    - Distal

- Expected Content (per artery): free-text, observational description such as -
    - "Flow restored"
    - "No residual narrowing seen"
    - "Not visualised"

5. **Stent Visualization** - Yes or no

SECTION 4: Report Validity and Correlation (for both pre-procedure stage and post-procedure stage in this package) ***(6 columns)***

4A. Pre-Procedure Report Validity

1. **Intra-Report Consistency Check:** Whether the written report's observations and impression agree with each other; expected values –

    - Consistent
    - Partially supported
    - Unsupported
2. **Reason (intra):** Free text (e.g. "Observations and impression match")

3. **Image vs Written Report Correlation (Inter):** Whether image features support what the written report claims; expected values –
    - Consistent
    - Partially supported
    - Unsupported
4. **Reason (inter):** Free text explaining mismatch or support

4B. Post-procedure Report Validity *(only inter-report validity in post-procedure stage required for this package)*

5. **Image vs Written Report Correlation (Inter):** Whether image features support what the written report claims; expected values –
    - Consistent
    - Partially supported
    - Unsupported
6. **Reason (inter):** Free text explaining mismatch or support

SECTION 5: Claim, Package & STG Alignment (overall) ***(4 columns)***

5A. Claim and Package Alignment

1. **Alignment Category:** Expected values -
    - Both pre-procedure and post-procedure investigations align
    - Only pre-procedure investigations align
    - Only post-procedure investigations align
    - None of the investigations align
2. **Reason for Alignment / Non-Alignment:** Free text (e.g. "Post-procedure angiogram mandatory as per STG")

5B. Investigation and STG Alignment

3. **STG Alignment Outcome:** Expected values -
    - Consistent
    - Partially supported
    - Unsupported
4. **Reason for Deviation (mapped to PPD / CPD logic):** Free text (e.g. "Flow obstruction as per STGs do not require procedure")

SECTION 6: Stage & Timeline Awareness ***(6 columns)***

S. No.	Column	Description
1	**Admission Date**	DD/MM/YYYY
2	**Pre-Procedure Investigation Date**	DD/MM/YYYY
3	**Post-Procedure Investigation Date**	DD/MM/YYYY
4	**Discharge Date**	DD/MM/YYYY
5	**Flag for one-month older documents for pre-procedure investigations**	Yes / No
6	**Flag before admission date or after discharge date documents for post-procedure investigations**	Yes / No

SECTION 7: Quality ***(1 column)***

- **Description:** free text, image usability comment, examples may include:
    - Good enough for human eyes
    - Poor quality
    - Artefacts present

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Summary PDF

- **Description:** Short human-readable case summary
- **Expected Content:**
    - Key image description summary
    - Report validity and correlation outcome
    - Package and STG alignment statement
    - Stage and timelines awareness summary of dates and flags
- **Tone:** Assistive, neutral, non-diagnostic

✅ **Important Rules for Participants**

- Use **observational language only**
- Do **not** confirm diagnoses (e.g., CAD, MI severity)
- Do **not** recommend approval or rejection
- Explicitly mention uncertainty if image quality is poor
- All fields must be populated (use "Not assessed / Not seen" where needed)